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Altered brain tissue microstructure and neurochemical profiles in long COVID and recovered COVID-19 individuals: A multimodal MRI study

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ABSTRACT

Background: Diverse neurological symptoms are experienced by long COVID and COVID-19 recovered individuals. However, the long-term effects of SARS-CoV-2 in the brain of both groups are underexplored. This study aimed to investigate changes in tissue microstructural and brain neurochemical levels in long COVID and recovered COVID-19 patients compared to healthy controls.
Methods: We recruited 47 participants (long COVID = 19, COVID-recovered healthy controls = 12, and healthy controls without COVID-19 infection = 16) who underwent 3T MRI scans. We acquired T1 and T2 weighted images to assess myelin signal, diffusion weighted images to assess tissue microstructure, and magnetic resonance spectroscopy data to estimate brain neurochemical levels.
Findings: Our multimodal MRI study showed altered T1w/T2w signal between long COVID vs COVID-recovered-healthy controls, long COVID vs healthy controls, and COVID-recovered-healthy controls vs healthy controls. Furthermore, T1w/T2w signal intensity was significantly correlated with physical and cognitive function. Diffusion weighted imaging also showed altered tissue microstructure in these three group comparisons. However, brain neurochemicals were only significantly different between long COVID vs COVID-recovered-healthy controls.
Interpretation: This is one of the first studies to report different myelin signal and brain neurochemical changes between long COVID, COVID-recovered-healthy controls, and healthy controls without SARS-CoV-2 infection. These brain changes provide compelling evidence for the long-term effects of SARS-CoV-2 on brain function.

1. Introduction

The infection caused by the SARS-CoV-2 (COVID-19) has had widespread global effects, ranging from mild and moderate symptoms to long-term health complications. While the majority of individuals with COVID-19 recover fully, approximately 10 % develop persistent symptoms referred to as long COVID (Centers for Disease Control and, 2022). According to the World Health Organization (WHO), long COVID is defined as the continuation or emergence of new symptoms in the three months following the initial SARS-CoV-2 infection, with these symptoms persisting for at least two consecutive months and not attributable to any other diagnosis (Soriano et al., 2022).

Emerging research indicates that people with long COVID commonly report a range of persistent symptoms, including fatigue, post-exertional malaise, sleep disturbance, and cognitive impairment (Soriano et al., 2022; Zhao et al., 2024a; Davis et al., 2023). Among these, cognitive impairment has emerged as a prominent neurological manifestation. A multi-centre cross-sectional study demonstrated that patients with post-COVID-19 conditions exhibit pronounced cognitive slowing, highlighting the substantial neurological impact of long COVID (Zhao et al., 2024a). Importantly, cognitive impairments are not limited to those with ongoing systemic symptoms; they are also evident in COVID-19 recovered individuals (COVID-RHC) who no longer report active symptoms, compared to healthy controls without SARS-CoV-2 infection (Non-COVID-HC) (Hampshire et al., 2021). Studies of COV-RHC have identified deficits in key cognitive domains, including processing speed, executive function, memory encoding and recall, with effects observed up to seven months post-infection (Hampshire et al., 2021). Collectively, these findings suggest central nervous system (CNS) involvement in both post-acute and long-term phases of SARS-CoV-2 infection, with measurable effects on cognitive functioning.

To investigate the underlying mechanism of cognitive impairments,

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magnetic resonance imaging (MRI) has been widely utilised to investigate brain structural, functional, and tissue microstructural changes in individuals with long COVID who have recovered from SARS-CoV-2 infection (Thapaliya et al., 2023, 2025; Nelson et al., 2024; Mishra et al., 2024; Petersen et al., 2023; Dadsena et al., 2025). In long COVID patients, MRI studies have consistently shown altered brain structures, including variations in brainstem, hippocampus, cerebellar volumes, and cortical thickness compared to healthy controls (Thapaliya et al., 2023, 2025; Pacheco-Jaime et al., 2025). Furthermore, impaired functional connectivity has been observed within the brainstem and between key intrinsic brain networks specifically, the salience network and the default mode network (Barnden et al., 2023; Li et al., 2023). Furthermore, magnetic resonance spectroscopy (MRS) has identified neurochemical alterations in long COVID patients demonstrating elevated levels of brain neurochemicals such as glutamate and N-acetyl-aspartate (Thapaliya et al., 2024). Structural MRI studies have also identified alterations in brain morphology among COVID-RHC, including reductions in grey matter volume, thickness and global brain volume when compared with healthy controls (Douaud et al., 2022; Lu et al., 2020). Diffusion-weighted imaging (DWI) has further revealed disruptions in white matter integrity in COVID-RHC (Mishra et al., 2024; Petersen et al., 2023; Nelson et al., 2025).

Despite the growing body of MRI-based research, gaps remain in our understanding of the full spectrum of neurological consequences of SARS-CoV-2 infection related to the discrepancies in the MRI findings (Nelson et al., 2025; Besteher et al., 2022; Serrano Del Pueblo et al., 2024). To date, no studies have directly compared patterns of myelin signal intensity, tissue microstructural and brain neurochemical levels changes across three distinct cohorts: individuals with long COVID, COVID-RHC, and Non-COVID-HC. Moreover, the T1-weighted/T2-weighted (T1w/T2w) whole brain myelin mapping technique (a sensitive MRI-based technique for assessing myelin signal intensity) (Ganzetti et al., 2015) has not yet been utilised to investigate potential differences in myelin signal across these groups. Therefore, this study aimed to assess myelin signal intensity using T1w/T2w ratio maps, tissue microstructural changes via diffusion-weighted imaging (DWI), and brain neurochemical levels estimated with magnetic resonance spectroscopy (MRS) in individuals with long COVID, COVID-RHC, and Non-COVID-HC. This comprehensive multimodal MRI approach facilitates a deeper understanding of the neurological impact of SARS-CoV-2 infection and provides deeper understanding of the brain damage related to cognitive impairment in both post-acute and long-term phases of the disease.

2. Methods

2.1. Participant recruitment

The study was approved by the Griffith University Human Research Ethics Committee (Ref: 2022/666) and conducted in accordance with the principles outlined in the Declaration of Helsinki. Written informed consent was obtained from all participants prior to inclusion in the study.

This cross-sectional study was carried out at the National Centre for Neuroimmunology and Emerging Diseases (NCNED), located on the Gold Coast, Queensland, Australia. Participants were recruited via the NCNED research registry database, as previously described in Thapaliya et al. (2025). Participants with long COVID were included based on the World Health Organization (WHO) working case definition (Soriano et al., 2022), which requires symptom onset within three months of confirmed or suspected SARS-CoV-2 infection and persistence of symptoms for at least two months. Healthy controls were individuals without a history of chronic illness or current medical conditions. All participants were aged between 18 and 65 years. A detailed medical history was collected to exclude individuals with comorbidities such as mental illness, malignancies, autoimmune disorders, neurological diseases, or cardiovascular conditions. Female participants who were pregnant or breastfeeding were excluded. A total of 19 individuals with long COVID and 27 healthy controls were enrolled. Among the healthy controls, 16 had no history of SARS-CoV-2 infection, while 12 were previously infected but had fully recovered and did not exhibit symptoms consistent with long COVID. Information regarding prior SARS-CoV-2 infection was collected using an online questionnaire via a Research Registry questionnaire developed by NCNED with the Centres for Disease Control and Prevention (CDC) and distributed online through Lime Survey and Redcap.

The severity measures for 'Duration of illness', 'Pain', and 'Physical function' were extracted from SF36v2 (Alonso et al., 1995). The severity of 'Pain' and 'Physical function' was assessed on a 0 to 100-point scale, 0 being very severe and 100 being no symptom. We also extracted a 'cognitive impairment' score from WHODAS [WHO Disability Assessment Schedule] version 2.0 (Üstün et al., 2010) scored on a 0-100-point scale, with higher scores indicating greater impairment (0 = no symptom, 100 = very severe), Dr Bell's Chronic Fatigue and Immune Dysfunction Syndrome (CFIDS) Disability Scale (Bell, 1995), and Modified Fatigue Impact Scale (Multiple Sclerosis Council for Clinical Practice Guidelines, 1998). Participants' demographic details and clinical and symptom severity scores are provided in Tables 1 and 2.

3. Methods and materials

MRI data acquisition.

3.1. T1 weighted and T2 weighted

T1w and T2w MRI data were acquired using a 3 T S Prisma MRI scanner (Siemens Healthcare, Erlangen, Germany) equipped with a 64-channel head-neck coil (Nova Medical, Wilmington, USA). Three-dimensional T1w images were obtained using a magnetization-prepared rapid gradient-echo (MPRAGE) sequence, with the following parameters: repetition time (TR) = 2400 ms, echo time (TE) = 1.81 ms, flip angle = 8°, acquisition matrix = 224 × 224 × 208, and isotropic voxel resolution of 1 mm^3. Three-dimensional T2w images were acquired using Siemens' T2 SPACE (Sampling Perfection with Application optimised Contrasts using different flip angle Evolution) sequence, with TR = 3200 ms, TE = 563 ms, variable flip angle, matrix size = 256 × 256 × 208, and voxel dimensions of 0.88 mm × 0.88 mm × 0.9 mm.

Table 1
Demographic and clinical characteristics of individuals with long COVID, COVID-RHC and Non-COVID-HC. Superscripts a, b, and c are the p-values for long COVID vs COVID-RHC, COVID-RHC vs Non-COVID-HC, and long COVID vs Non-COVID-HC respectively. F=Female, M = male, COVID-RHC =COVID-19 recovered healthy controls, Non-COVID-HC = healthy controls without SARS-CoV-2 infection, MFS=Modified Fatigue Impact Scale.

	Long COVID (n = 19)	COVID-RHC (n = 12)	Non-COVID-HC (n = 16)	p-value
Diagnostic criteria	**WHO**	N/A	N/A	
Age	47.95 ± 13.28	34.76 ± 7.10	38.85 ± 12.53	0.012[a], 0.083[b], 1[c],
Sex (F/M)	14/5	9/3	11/5	N/A
Duration (years)	0.78 ± 0.69	N/A	N/A	N/A
Pain	45.88 ± 23.18	87.5 ± 18.5	89.21 ± 17.3	<0.001[a], 1[b], <0.001[c]
Cognitive	32.69 ± 19.24	8.18 ± 17.28	2.86 ± 5.4	0.004[a], 0.84[b], <0.001[c]
Physical function	59.17 ± 27.98	90.0 ± 27.38	97.19 ± 9.99	<0.014[a], 1[b], <0.001[c]
Bell disability score	51.25 ± 20.62	97.50 ± 8.66	98.57 ± 3.63	<0.001[a], 1[b], <0.001[c]
MFIS	59.06 ± 15.91	8.83 ± 16.10	5.67 ± 7.25	<0.001[a], 1[b], <0.001[c]

Table 2
Long COVID symptom severity scores obtained from self-reported questionnaire.

Severity	None	Very mild	Mild	moderate	severe	Very severe	Missing data
Long COVID							
Fatigue				58 %) (11/19)	42 % (8/19)		
Concentration	6 % (1/18)		22 % (4/18)	39 % (7/18)	33 % (6/18)		1
Muscle ache	5 % (1/19)	5 % (1/19)	26 % (5/19)	37 % (7/19)	26 % (5/19)		
Headaches	5 % (1/19)		26 % (5/19)	58 % (11/19)	11 % (2/19)		
Odour/taste	65 % (11/17)		6 % (1/17)	29 % (5/17)			2
Unrefreshing sleep	11 % (2/19)		5 % (1/19)	37 % (7/19)	42 % (8/19)	5 % (1/19)	
Sore throat	44 % (8/18)	6 % (1/18)	22 % (4/18)	17 % (3/18)	6 % (1/18)	6 % (1/18)	1
Muscle weakness	39 % (7/18)	16 % (3/18)	22 % (4/18)	1/18 (6 %)	16 % (3/18)		1
Depression	50 % (5/10)		30 % (3/10)	10 % (1/10)	10 % (1/10)		9

The total scan durations were 8 min and 20 s for T1w imaging and 5 min and 44 s for T2w imaging. All MR images were acquired for the three groups long COVID COVID-RHC, and Non-COVID-HC using the same scanner and identical scanning protocols.

3.2. Diffusion tensor imaging

The diffusion data were acquired using a 3T Prisma MRI scanner (Siemens Healthcare, Erlangen, Germany) with a 64-channel head–neck coil (Nova Medical, Wilmington, USA). DTI data were acquired using 2-shell acquisition protocols: 30 directions at $b = 1000$ s/mm^2 and 66 directions at b = 2500 s/mm^2, along with nine b = 0 scans. Other settings were repetition time/echo time = 4100/75 ms, field of view (FOV) = 244 × 244, matrix = 122 × 122, voxel dimension of 2.0 mm^3 and 66 slices. The diffusion parameters were estimated using only b-values of 1000 s/mm^2.

3.3. Magnetic resonance spectroscopy

Magnetic resonance spectroscopy (MRS) data were acquired as in our previous publications (Thapaliya et al., 2024) for all participants using a 3 T S Prisma MRI scanner (Siemens Healthcare, Erlangen, Germany) equipped with a 64-channel receive-only head coil. In brief: MRS data were collected from a manually placed single voxel (20 × 20 × 20 mm^3) in the posterior cingulate cortex (PCC) guided by the T1w image acquired as above.

Magnetic field homogeneity was optimised using gradient echo shimming (first- and second order) and FASTMAP-based fine adjustment of first-order shims (Shah et al., 2009). Spectra data was acquired using a stimulated echo acquisition mode (STEAM) pulse sequence (TE = 6.5 ms, mixing time (Tm) = 75 ms, TR = 4000 ms, and 128 signal averages) with water suppression via variable power radiofrequency pulses with optimised relaxation delays. Unsuppressed water spectra were also acquired for quantification purposes. All MRS data were exported in RDA format for preprocessing and analysis.

4. MRI data processing

4.1. T1w/T2w whole brain myelin mapping

T1w and T2w MRI data were pre-processed using the workflow described by (Ganzetti et al., 2015) to generate whole-brain T1w/T2w ratio maps. The preprocessing pipeline includes co-registration of the T2w image to the T1w image using a rigid-body transformation (Collignon et al., 1995) to ensure anatomical alignment. Bias field correction was then performed separately on both T1w and T2w images using the algorithm implemented in SPM12 (Ashburner and Friston, 2005; Weiskopf et al., 2011). The parameters for smoothing and regularisation within the bias correction procedure were maintained at their default values as specified by the SPM12 software. Following bias correction, the intensity of both T1w and T2w images was standardised using a linear scaling method adapted from (Ganzetti et al., 2015). After normalisation, voxel-wise T1w/T2w ratio maps were computed.

4.2. DTI data processing

Diffusion-weighted data were denoised using the 'dwidenoise' command from MRtrix3 (https://www.mrtrix.org/), which estimates spatially varying noise levels using an algorithm based on random matrix theory (Cordero-Grande et al., 2019). Eddy current induced distortions and subject motion between and within diffusion-weighted imaging (DWI) volumes were corrected using the 'dwipreproc' command in MRtrix3, which performs affine registration of each DWI volume to a reference b = 0 image. Diffusion tensor metrics including fractional anisotropy (FA), axial diffusivity (AD), radial diffusivity (RD), and mean diffusivity (MD) were calculated using the FMRIB Diffusion Toolbox (FDT), part of the FMRIB Software Library (FSL) (Smith et al., 2004). The 'dtifit' tool was used to fit the diffusion tensor model and generate parameter maps for each subject. For voxelwise group analysis, each subject's FA map was nonlinearly registered to the Montreal Neurological Institute (MNI) standard space (FSL_HCP1065_FA_1 mm) using the Tract-Based Spatial Statistics (TBSS) pipeline in FSL. Non-FA images (i.e., AD, RD, and MD) were warped into standard space using the deformation fields obtained from the FA-based co-registration.

4.3. MRS data processing

Spectral data were post-processed using the Osprey software package (Oeltzschner et al., 2020). Raw MRS data were first imported into Osprey and underwent the following preprocessing steps: eddy current correction, frequency and phase alignment across transients, water signal suppression, frequency referencing, and initial phasing. These operations were performed using the 'OSpreyProcess' module (Oeltzschner et al., 2020).

Post-processed spectra were subsequently fitted using the Linear Combination Modeling (LCModel) algorithm (Provencher, 1993) over a spectral range of 0.5–4.5 ppm, with a knot spacing of 5. A TE = 0 basis set was selected for metabolite fitting, as recommended for STEAM sequences with echo times less than 11 ms. The following metabolites were quantified: Alanine (Ala), Aspartate (Asp), Creatine (Cr), Phosphocreatine (PCr), γ-Aminobutyric acid (GABA), Glucose (Glc), Glutamine (Gln), Glutamate (Glu), Glycerophosphocholine (GPC), Phosphocholine (PCh), Glutathione (GSH), Myo-inositol (Ins), Lactate (Lac), N-Acetylaspartate (NAA), N-Acetylaspartylglutamate (NAAG), Scyllo-inositol (Scyllo), and Taurine (Tau).

All fitted spectra were visually inspected for artifacts or poor model fit. Spectral quality was assessed by determining the linewidth (full width at half maximum, FWHM) of the creatine (Cr) peak via a single Lorentzian fit. The mean linewidth across participants was 5.63 ± 0.55 Hz. Signal-to-noise ratio (SNR) was calculated as the ratio of the Cr peak amplitude to the standard deviation of detrended noise in the spectral region between −2 and 0 ppm, yielding an average SNR of 50 ± 6.

Cramér–Rao lower bounds (CRLBs), an estimate of the uncertainty in metabolite concentration fits, were computed within Osprey for each

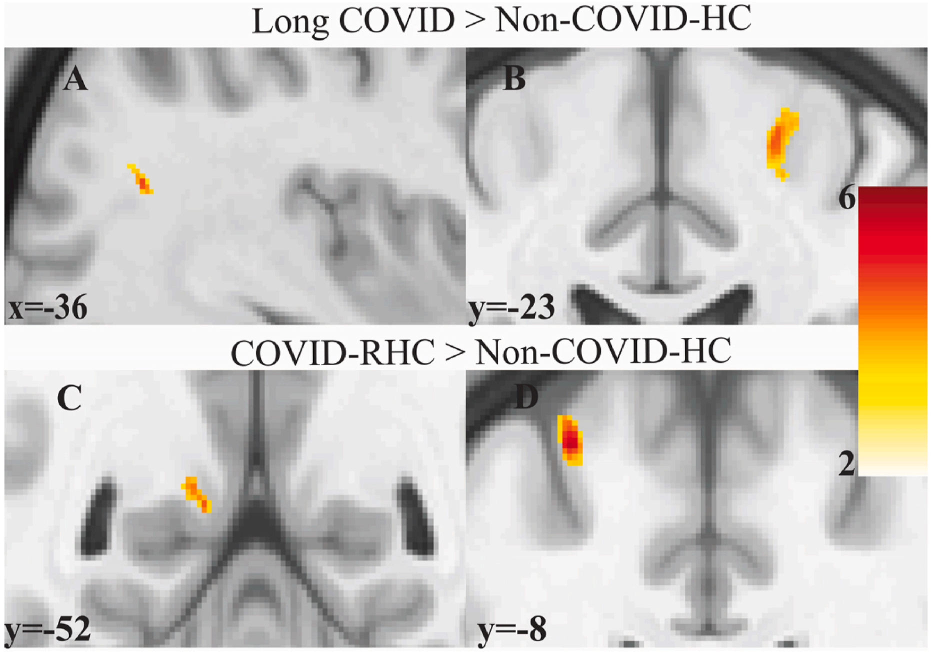

Fig. 1. Clusters where T1w/T2w signal intensity differed between the long COVID, healthy controls and recovered healthy controls. T1w/T2w signal intensity is significantly higher A) in the middle temporal gyrus; B) in the precentral gyrus in long COVID compared with healthy controls with no previous COVID-19 infection (Non-COVID-HC). T1w/T2w signal intensity was significantly higher in C) posterior cingulate cortex and D) Middle frontal gyrus in COVID-RHC compared to Non-COVID-HC. Voxels from the SPM group comparison (red-yellow) have $T > 2$. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

quantified metabolite (Gyngell et al., 1991). To ensure accurate anatomical localisation, the MRS voxel information (position, size, and orientation) was used to generate a binary mask, which was coregistered to the individual's T1-weighted MPRAGE image using the 'Coreg' module in Osprey. This ensured precise alignment of the spectroscopic voxel with underlying brain anatomy.

The MPRAGE image was segmented into grey matter, white matter, and cerebrospinal fluid (CSF) tissue classes using the segmentation tool in SPM12. The coregistered MRS voxel mask was then overlaid onto the segmented tissue maps. Fractional tissue volumes (grey matter, white matter, CSF) within the MRS voxel were estimated using the 'Seg' module in Osprey (Oeltzschner et al., 2020).

Absolute metabolite concentrations in the posterior cingulate cortex (PCC) were estimated by correcting for tissue-specific relaxation effects using the method described by Gasparovic et al. (2006).

4.4. Statistical analysis

Voxel based statistical comparison was performed similar to our previous findings (Thapaliya et al., 2020, 2021). Analysis was performed across all three groups long COVID, COVID-RHC and Non-COVID-HC for T1w/T2w ratio maps, FA, AD, MD, and RD using SPM12 software. Voxel clusters were identified using a threshold of uncorrected voxel-wise $p < 0.001$ and a minimum cluster size of 60 voxels. Statistical significance was assessed using false discovery rate corrected cluster-level p-values (cluster p-FDR). Significant clusters were overlaid on a standard T1-weighted template (mni_icbm152_t1_tal_nlin_sym_09a). All analyses were adjusted for age and gender. Cluster locations were identified using the xjview toolbox (https://www.alivelearn.net/xjview) and FSL software(Smith et al., 2004).

Group comparisons of clinical characteristics and brain neurochemicals between long COVID, COVID-RHC and Non-COVID-HC were conducted using univariate general linear models is SPSS version 29 (IBM, Armonk, NY). Multiple comparisons were corrected using the False Discovery Rate. Age and sex were included as covariates in the analysis.

We did not perform brain neurochemical levels statistical test between long COVID and Non-COVID-HC because this has already been reported in our previous publication (Thapaliya et al., 2024).

5. Results

5.1. Clinical and symptom characteristics

The clinical symptom severity scores of long COVID patients are provided in Tables 1 and 2 The majority of participants were female, long COVID individuals (14/19), COVID-RHC (9/12), Non-COVID-HC (11/19). Participants completed the online questionnaire before visiting the MRI site. The long COVID individuals experienced a higher pain level (<0.001 vs COVID-RHC and Non-COVID-HC), impaired cognitive function (0.004 vs COVID-RHC; <0.001 vs Non-COVID-HC), impaired physical function (0.014 vs COVID-RHC; <0.001 vs Non-COVID-HC), and impaired Bell disability score and Modified Fatigue Impact Scale (<0.001 vs both groups). No significant differences were observed between COVID-RHC and Non-COVID-HC groups across any of these clinical domains (see Table 1).

Most of the long COVID participants reported complaints for fatigue (19/19, 100%); Unrefreshing sleep (16/19, 84%), concentration (13/18, 72%), headache (13/19, 69%), and muscle ache (12/19, 63%) ranging from moderate-to-very severe (see Table 1). In contrast, muscle weakness (14/18, 77%), Odour/taste (12/17, 71%), sore throat (13/18, 62%), and depression (8/10, 80%) ranging from none to mild severity were less common in long COVID individuals (see Table 2).

5.2. T1w/T2w mapping

We estimated the whole-brain T1w/T2w map in long COVID, COVID-RHC, and Non-COVID-HC.

5.2.1. Long COVID vs non-COVID-HC

Voxel-based group comparisons revealed significantly increased T1w/T2w signal in the precentral gyrus (*pfdr* < 0.001) and the middle temporal gyrus (*pfdr* = 0.005) in long COVID compared to healthy controls (see Fig. 1).

5.2.2. Long COVID vs COVID-RHC

T1w/T2w signals were significantly increased in COVID-RHC when compared to long COVID in multiple brain regions (see Fig. 2, Table 3). The most prominent increases were observed in the pons (*pfdr* < 0.001), left and right midbrain (*pfdr* < 0.001) and cerebellar tonsil (*pfdr* < 0.001). Additional regions showing significant differences between two groups were left superior longitudinal fasciculus (*pfdr* < 0.001), precentral gyrus (*pfdr* = 0.009), superior longitudinal fasciculus (*pfdr* = 0.007), forceps minor (*pfdr* = 0.005), right optic radiation (*pfdr* = 0.017), and Brodmann area 18 (*pfdr* = 0.005).

5.2.3. COVID-RHC vs non-COVID-HC

We observed significantly increased T1w/T2w signal intensity in the precentral gyrus (*pfdr* = 0.001) and the posterior cingulate cortex (*pfdr* = 0.004) in COVID-RHC compared to Non-COVID-HC (see Fig. 1).

5.3. Diffusion tensor imaging

5.3.1. Long COVID vs non-COVID-HC

The mean diffusivity was significantly lower in the dentate regions (pfdr = 0.041, X = 14, Y = −62, and Z = −28, cluster size = 91) in long COVID compared to Non-COVID-HC (see Fig. 3).

5.3.2. Long COVID vs COVID-RHC

FA value was significantly higher in right SLF (pfdr = 0.001, X = 37, Y = −35, and Z = 37, cluster size = 125) in long COVID compared to COVID-RHC (see Fig. 3).

5.3.3. COVID-RHC vs non-COVID-HC

AD, MD, and RD values were significantly lower in left caudate (AD: *pfdr* = 0.001, X = −16, Y = 24, Z = −1, cluster size = 105; MD: *pfdr* = 0.005, X = −16, Y = 24, Z = −2, cluster size = 118; RD: *pfdr* = 0.020, X = −16, Y = 24, Z = −3, cluster size = 100) in COVID-RHC compared to Non-COVID-HC (see Fig. 4). Axial diffusivity was also significantly lower in the supramarginal gyrus (*pfdr* = 0.008, X = 36, Y = −22, Z = 31, cluster size = 66) in COVID-RHC compared to Non-COVID-HC.

5.4. Magnetic resonance spectroscopy

5.4.1. Long COVID vs COVID-RHC

MRS analysis showed only glutamine was significantly higher in COVID-RHC (5.612 ± 0.505, *pfdr* = 0.019) compared to long COVID (5.358 ± 0.569) after adjusting for multiple comparisons. Conversely, NAA was significantly higher in long COVID (*pfdr* = 0.009, 15.041 ± 0.525) compared to COVID-RHC (14.525 ± 0.420) after adjusting for multiple comparisons (see Fig. 5). Other neurochemicals that were not significantly different after adjusting multiple comparisons are shown in Fig. 6.

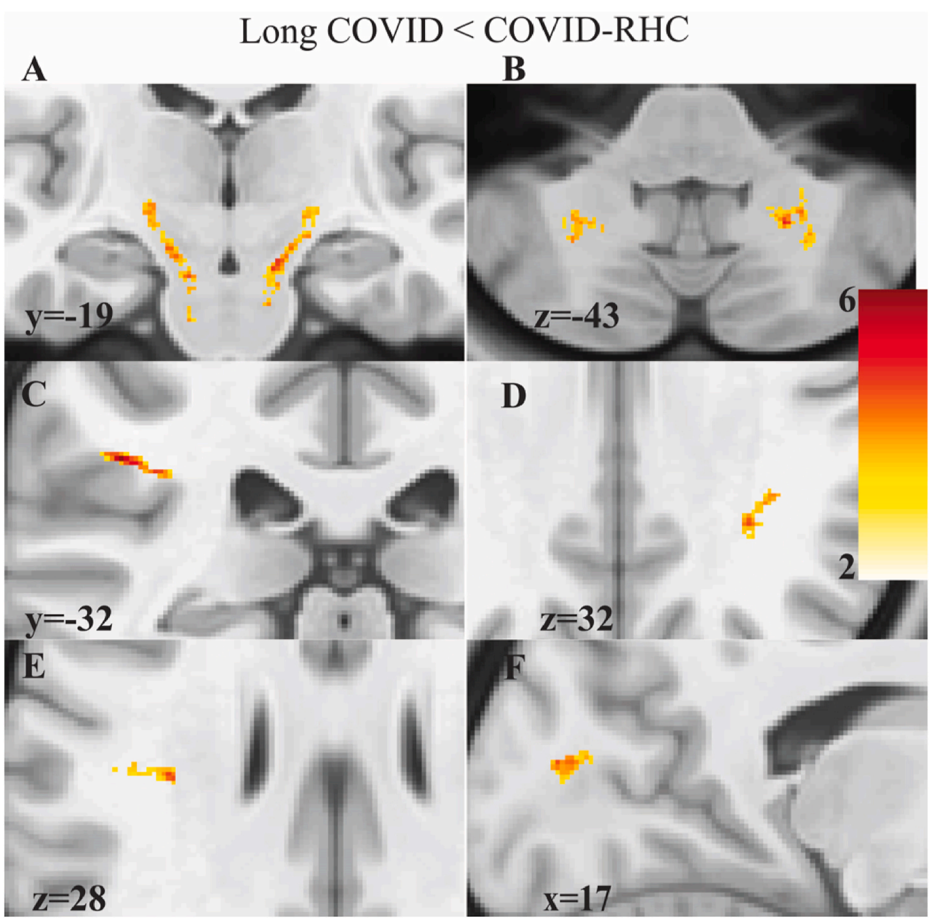

Fig. 2. T1w/T2w signal intensity was significantly higher in COVID-19 recovered healthy controls (COVID-RHC) compared to long COVID across various brain regions. A) Pons/midbrain, B) Cerebellar Tonsil, C) Left SLF, D) Right SLF, E) Left SLF, and F) Brodmann area 18. Voxels from the SPM group comparison (red - yellow) have T > 2. SLF = superior longitudinal fasciculus. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 3
Cluster statistics of T1w/T2w group comparisons between COVID-RHC, HC, and long COVID. The voxel P threshold for cluster formation was 0.001. SLF=Superior longitudinal fasciculus, COVID-RHC= COVID-19 recovered individuals, Non-COVID-HC healthy controls without COVID-19 infection, fdr = false discovery rate.

Group	Peak X Y Z	Area	Cluster size (voxels)	Cluster p-fdr
Long COVID > Non-COVID-HC	25 -23 60	Precentral Gyrus	132	<0.001
	−36 -66 21	Middle Temporal Gyrus	72	0.005
COVID-RHC > Long COVID	−47 -31 28	Left SLF (ROI3)	242	<0.001
	−8 -18 -28	Pons (ROI4)	285	<0.001
	14 -20 -14	Midbrain (ROI5)	256	<0.001
	22 -55 -43	Cerebellar Tonsil (ROI6)	145	<0.001
	−25 -9 58	Precentral gyrus (ROI7)	72	0.009
	−34 -19 28	Left SLF (ROI8)	76	0.007
	31 -42 32	Right SLF (ROI9)	81	0.005
	17 -85 13	Brodmann area 18 (ROI10)	83	0.005
	20 -68 -1	Right optic radiation	63	0.017
	12 52 -13	Forceps minor (ROI11)	83	0.005
	−17 -20 -6	Midbrain	94	0.003
	−23 -54 -44	Cerebellar Tonsil	89	0.004
COVID-RHC > Non-COVID-HC	−27 -8 63	Precentral gyrus (ROI12)	131	0.001
	−10 -52 9	Posterior Cingulate (ROI13)	91	0.004

5.4.2. COVID-RHC vs non-COVID-HC

The choline compound was significantly higher in COVID-RHC compared to long COVID. But this significance did not survive the adjustment for multiple comparisons (see Fig. 6).

5.5. Correlations with clinical measures

In long COVID, we observed a positive correlation between T1w/T2w signal intensity and physical function ($p = 0.024$, $r = 0.56$, see Fig. 7A) in the middle temporal gyrus. Additionally, we also observed negative correlations between T1w/T2w signal intensity and cognitive scores ($p = 0.01$, $r = -0.64$, see Fig. 7B) in the midbrain.

No significant correlations were observed between any DTI parameters and clinical measures in long COVID. These non-significant results are provided for reference in Supplementary Table 3.

6. Discussion

Our study identified altered T1w/T2w signal, tissue microstructure, and brain neurochemical levels among individuals with long COVID, COVID-RHC, and Non-COVID-HC using multi-modal MRI approach. These novel findings suggest that COVID-19 may lead to lasting brain alterations that potentially impact overall brain function even after recovery.

6.1. Long COVID vs non-COVID-HC

The T1w/T2w myelin mapping showed changes in the precentral gyrus and middle temporal gyrus in long COVID patients. The T1w/T2w technique has been shown to map myelin signal intensity (Ganzetti et al., 2015). The significant increases in the T1w/T2w signal intensity in the precentral and middle temporal gyrus indicates relatively higher myelin signal in long COVID patients compared to Non-COVID-HC. The precentral gyrus contains a high density of myelin and regulates motor function (Dubbioso et al., 2021) which is impaired in long COVID (Besteher et al., 2022). Similarly, the middle temporal gyrus plays a critical role in memory and cognitive processes (Japee et al., 2015). However, our findings contradict the early post-mortem study of brain tissue of the nine humans who died after COVID-19 which demonstrated persistent loss of myelinated axons compared to healthy controls (Fernández-Castañeda et al., 2022) at the acute phase of the disease (Fernández-Castañeda et al., 2022). Our findings of higher T1w/T2w signal in long COVID potentially indicates re-myelination necessary for brain-body communication. This is supported by an animal viral model study in multiple sclerosis demonstrating spontaneous remyelination following extensive demyelination (Murray et al., 2001). Our previous study in Myalgic Encephalomyelitis (ME), a condition with symptoms overlapping those of long COVID (Weigel et al., 2024) also revealed significantly elevated T1w/T2w signal intensity (Thapaliya et al., 2020). Therefore, there is the possibility of remyelination in individuals living with long COVID for the proper functioning of the central nervous

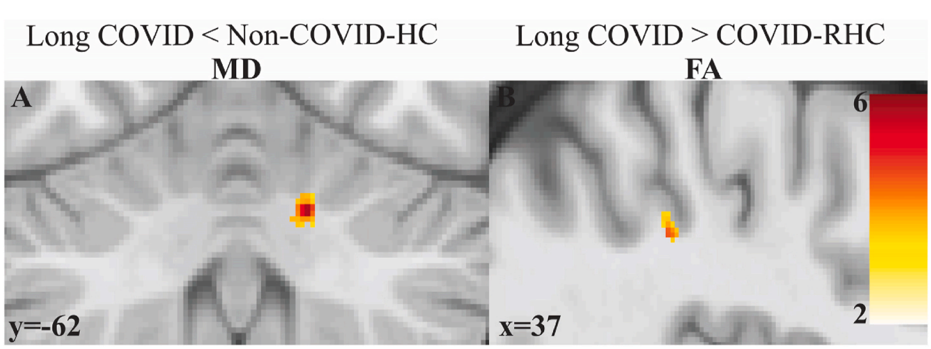

Fig. 3. Shows clusters with significant differences in mean diffusivity and fractional anisotropy. A) significant lower mean diffusivity in long COVID compared to Non-COVID-HC in the dentate regions. B) significantly higher fractional anisotropy in long COVID compared to COVID-RHC in the right superior longitudinal fasciculus. Voxels from the SPM group comparison (red-yellow) have T > 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

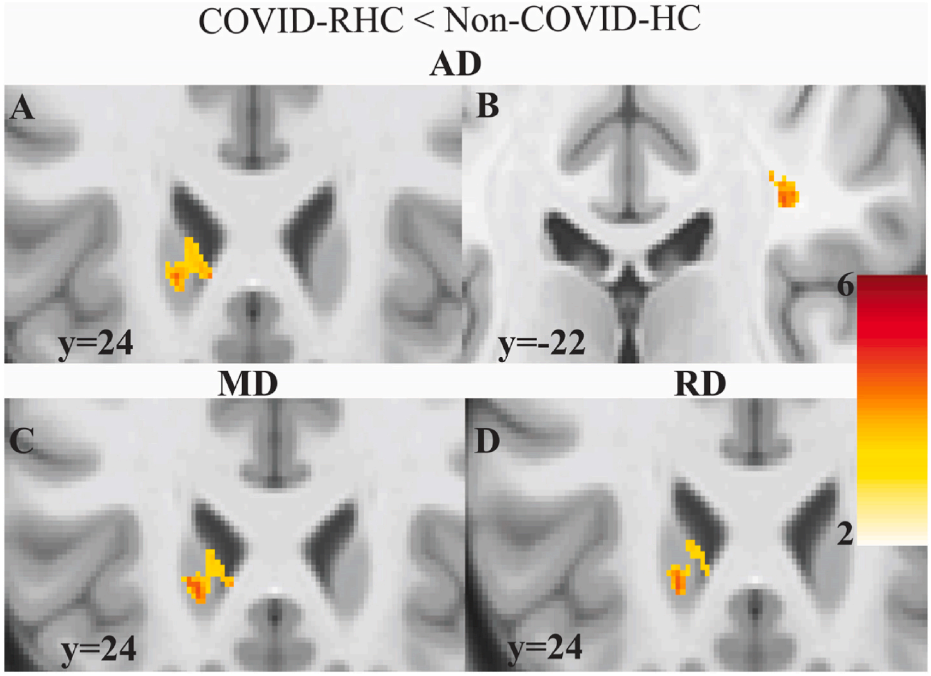

Fig. 4. Clusters for different AD, MD and RD in COVID-RHC and Non-COVID-HC. A), C), and D) demonstrate significantly lower axial, mean, and radial diffusivity values in the left caudate region of COVID-RHC compared to Non-COVID-HC. B) shows significantly reduced AD in COVID-RHC compared to Non-COVID-HC in the right supramarginal gyrus.

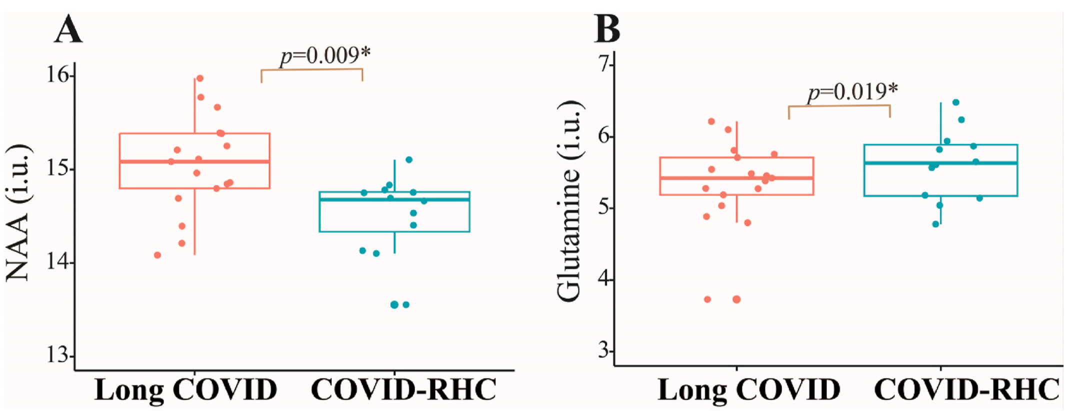

Fig. 5. Box plots of brain neurochemicals estimated in long COVID and COVID-RHC. A) N-acetyl-aspartate level was significantly higher in long COVID compared to COVID-RHC. B) glutamine level was significantly higher in COVID-RHC compared to long COVID. i. u. = arbitrary unit. The rectangular box represents the first quartile (Q1, 25th percentile) to the third quartile (Q3, 75th percentile), representing the middle 50 % of the data. The line inside the box marks the median (Q2), which is midpoint of the dataset. Lines extend from each end of the box to the minimum and maximum data points that are not considered outliers, indicating the data's spread.

system. It is not known whether increased T1w/T2w signal may also indicate inflammation or gliosis in long COVID patients (Chen et al., 2025).

Our findings of lower mean diffusivity in long COVID when compared with healthy controls were similar to the previous findings (Petersen et al., 2023; Lu et al., 2020; Díez-Cirarda et al., 2023). Our previous findings in ME also showed lower MD values in the brainstem compared to healthy controls (Thapaliya et al., 2021). The decreased MD values suggested a limited diffusion, indicating a possible intrinsic re-construction (e.g. remyelination) process that occurred after infection (Cauley and Cataltepe, 2014).

6.2. Long COVID vs COVID-RHC

The T1w/T2w signal intensity was significantly elevated in COVID-RHC compared to long COVID in different brain regions. The most prominent differences occurred in the brainstem, superior longitudinal fasciculus, and cerebellum regions. The brainstem has reticular activation nuclei that influence cortical function (Benarroch, 2018). Furthermore, these nuclei generate oscillatory signals that facilitate the coherence of cortical oscillation necessary for attention, sleep-wake cycle, pain and respiration, problem solving, and memory (Garcia-Rill et al., 2013). Impairment of these functions were experienced by COVID-RHC which were slower in performing a cognitive task (Hampshire et al., 2021). The altered signal in the superior longitudinal fasciculus could affect the performance of language, memory and motor function because it contains the major white matter tracts that connect the different brain regions involved (Cai et al., 2019). Additionally, the cerebellum has the largest number of neurons responsible for coordinating motor movement and maintaining balance and posture

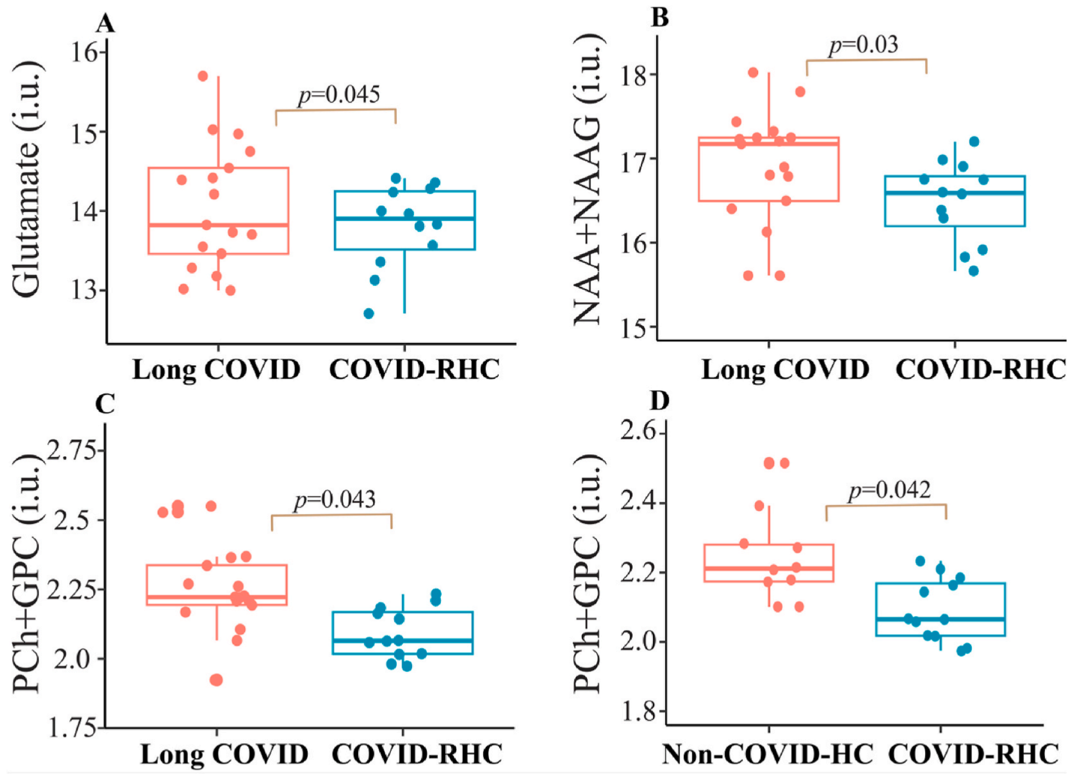

Fig. 6. Shows brain neurochemicals estimated in long COVID, COVID-RHC, Non-COVID-HC that were not significantly different after adjusting for multiple comparisons. A) Glutamate, B) N-acetyl-aspartate and N-acetyl-aspartyl-glutamate (NAA + NAAG), C) phosphorylcholine and glycerylphosphorylcholine (PCH + GPC) levels was higher in long COVID compared to COVID-RHC. D) PCh + GPC were higher in Non-COVID-HC compared to COVID-RHC. i. u. = arbitrary unit.

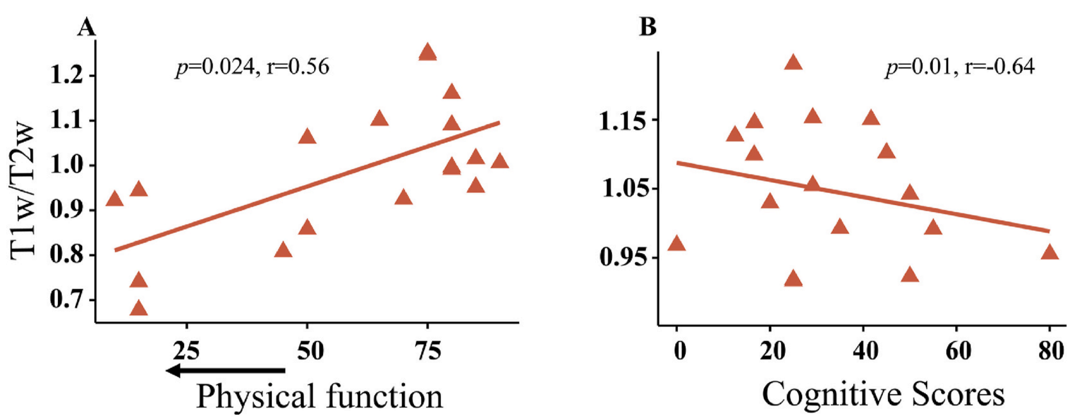

Fig. 7. Correlation plots between T1w/T2w and severity measures in long COVID. A) T1w/T2w signal intensity in the Middle temporal gyrus vs Physical function. B) T1w/T2w signal intensity in the Midbrain vs cognitive dysfunction. Y-axis is the T1w/T2w signal intensity values, X-axis is the clinical measures. Lower physical function (shown by arrow) is greater impairment in long COVID. Lower cognitive score indicates more impaired cognition.

(Glickstein and Doron, 2008). Therefore, altered signal in cerebellum regions may impact the physical functions which are most impaired in long COVID and COVID-RHC (Hampshire et al., 2021). However, we were unable to compare our T1w/T2w signal intensity (relative myelin signal) findings with previous studies due to the lack of existing research. However, a volumetric study reported reduced cortical thickness and altered white matter integrity in long COVID patients compared to COVID-RHC (Serrano Del Pueblo et al., 2024).

Additionally, we observed higher FA values in the supramarginal gyrus of long COVID patients. Similar increases in FA have been reported in the thalamic regions in long COVID (Heine et al., 2023). However, other studies reported lower FA values in the white matter regions of long COVID when compared with COVID-19 recovered healthy controls (Pacheco-Jaime et al., 2025; Serrano Del Pueblo et al., 2024). These inconsistencies may be attributed to differences in the diffusion data acquisition protocols and varying MRI field strengths (Zhan et al., 2013). Another study reported both increased and decreased FA values in COVID-19 recovered healthy controls (Mishra et al., 2024), further highlighting the variability in findings. These discrepancies demand new longitudinal and multi-centre studies. Longitudinal studies in both long COVID and those with COVID-19 recovered healthy controls could also help to clarify how FA changes relate to time since infection.

We also found significant alterations in brain neurochemical levels in both long COVID and COVID-RHC. Specifically, individuals with long COVID showed a significant reduction in glutamine levels suggesting increased utilisation of glutamine. This finding aligns with previous studies (López-Hernández et al., 2023; Saito et al., 2024), where during the COVID-19 acute phase, a decrease in circulating levels of glutamine has been widely described (Wu et al., 2021). This depletion is thought to result from glutamine being consumed for Adenosine triphosphate production and immune cell regulation (López-Hernández et al., 2023). Conversely, we found a significant increase in NAA levels in long COVID. This elevation may reflect a compensatory response to reduced brain metabolic efficiency, potentially involving increased glucose and oxygen utilisation. Additionally, osmotic stress triggered by COVID-19 symptoms may contribute to elevated NAA levels (Chiu et al., 2014; Weissenborn et al., 2004).

6.3. COVID-RHC vs non-COVID-HC

The T1w/T2w signal intensity was significantly higher in the precentral gyrus and the posterior cingulate cortex in the COVID-RHC compared to Non-COVID-HC. This increase may reflect a re-myelination process, similar to that observed in long COVID when compared with Non-COVID-HC, as previously discussed. Currently, there are no other studies reporting T1w/T2w signal changes in these cohorts. However one study reported increased volume in the cingulate gyrus of COVID-19 survivors after a 3-month follow-up study (Lu et al., 2020). In contrast, a UK biobank follow-up study comparing individuals before and after COVID-19 infection found decreased global grey matter volume in the COVID-RHC (Douaud et al., 2022). Additionally, a resting-state fMRI 1-year follow-up study found that COVID-19 survivors showed altered amplitudes of low-frequency fluctuation in the left precentral gyrus, angular gyrus, and thalamus (Zhao et al., 2024b). Since, precentral gyrus and posterior cingulate cortex play a significant role in motor performance and cognition (Bussey et al., 1997), the impairments in these regions may contribute to the cognitive problem in COVID-19 recovered healthy controls.

Furthermore, we found MD, AD, and RD values were significantly lower in the COVID-RHC compared to Non-COVID-HC in caudate regions which play a key role in motor execution and learning (Lanciego et al., 2012). Our findings support the previous findings where MD, AD, and RD values were reported lower in COVID-19 recovered healthy controls compared to healthy controls (Lu et al., 2020). Another study in COVID-recovered healthy controls found lower MD in the white matter tracts compared to healthy controls (Petersen et al., 2023). Similarly, another study reported reduced AD and RD in the cingulum hippocampus and arcuate fasciculus in COVID-19 recovered healthy controls (Mishra et al., 2024). Studies have found both increased and decreased FA, MD, AD, and RD in COVID-RHC compared to healthy controls in different brain regions(Mishra et al., 2024; Petersen et al., 2023; Huang et al., 2023). Altered FA, MD, AD, and RD in different brain regions may explain the effects of COVID-19 in diverse brain regions suggesting a massive microglial activation following SARS-Cov-2 infection, prompting neuroinflammation (Petersen et al., 2023). The cognitive impairment experienced by long COVID and COVID-RHC (Serrano Del Pueblo et al., 2024) could be due to the tissue microstructural changes demonstrated by our multi-modal MRI findings. The impaired tissue microstructure was also reported in individuals with long-COVID (Hosp et al., 2024). Therefore, the underlying mechanisms contributing to the cognitive complaints in long COVID and COVID-RHC are likely a combination of multiple factors, including viral neurotropism and widespread systemic inflammation (Serrano Del Pueblo et al., 2024).

6.4. Correlation between T1w/T2w and clinical measures

Our study found a significant positive correlation between T1w/T2w signal intensity in the middle temporal gyrus and physical function in long COVID, indicating lower physical activity is associated with decreased T1w/T2 signal intensity. This finding aligns with a preclinical study linking physical exercise with improved myelin health (Jensen et al., 2018). A recent animal study demonstrated physical activity enhances motor cortex myelination (Zheng et al., 2019). Similarly, in humans 24 weeks of aerobic exercise led to increased T1w/T2w signal intensity suggesting improved myelin integrity (Mendez et al., 2021). A study in healthy young and older individuals suggest that higher cardiorespiratory fitness is associated with higher myelin content (Faulkner et al., 2024). A study of cerebral small vessel disease showed that higher physical activity was associated with higher myelin content (Hainsworth et al., 2024). Thus, decreased T1w/T2w signal intensity (indicating lower myelin content) in long COVID could be due to the reduced physical activity resulting in more axons being demyelinated.

Additionally, we found a significant negative correlation between T1w/T2w signal intensity in the midbrain and cognitive function, indicating greater cognitive impairment is associated with decreased T1w/T2w signal intensity. Myelin is essential for efficient neural impulse conduction and supports rapid information processing in the brain (Connor and Menzies, 1996). Thus, reductions in the myelin content may directly contribute to cognitive dysfunction in long COVID. Previous studies have established the critical role of myelin in maintaining cognitive function and its disruption in neurodegenerative diseases (Khelfaoui et al., 2024). A study in healthy adults showed lower myelin content is associated with declined executive function (Gong et al., 2023) and also white matter myelination was associated with processing speed across the adult life span (Gong et al., 2024). Therefore, we suggest that reduced myelin content in long COVID could potentially affect brain connectivity patterns causing cognitive dysfunction.

Strengths of this work lie in its high-quality imaging data; a robust and reproducible image processing pipeline; and the investigation of change in myelin signal in long COVID and COVID-19 recovered healthy controls compared to healthy controls without previous COVID-19 infection. This is the first study identifying myelin signal in different groups potentially showing altered myelin signal in different brain regions. However, this study has several limitations. 1) This is a cross-sectional study, so the present work can only identify brain changes in long COVID, COVID-19 recovered healthy controls, and healthy controls without infection in a small sample size. 2) This is an exploratory study, and limited research has compared MRI findings between long COVID and COVID-19 recovered healthy controls. Due to the novelty of the disease, the purpose of these analyses was to generate new knowledge and hypotheses for future studies and needs to be interpreted with caution. Furthermore, the statistical thresholding used in this study might introduce Type I error. Therefore, to mitigate this, future studies must seek to replicate these findings in larger cohorts. Additionally, a longitudinal study is needed to investigate whether these findings are progressive or stable.

7. Conclusion

Our study identified altered signal intensity, abnormal tissue microstructure, and imbalanced neurochemicals in long COVID and COVID-19 recovered healthy controls. Importantly, we also found a significant association between T1w/T2w signal intensity and clinical measures, suggesting a potential link between myelin content and symptom severity. These findings provide essential insights into long-term neurological effects of COVID-19 impacting brain function in long COVID and COVID-19 recovered healthy controls. Therefore, continuous monitoring and tailored treatments may be necessary for these individuals. Furthermore, long-term follow-up studies are crucial to determine whether the identified brain changes are progressive, stable or dynamic over time.

CRediT authorship contribution statement

Kiran Thapaliya: Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Sonya Marshall-Gradisnik:** Writing – review & editing, Supervision, Funding acquisition. **Maira Inderyas:**

Writing – review & editing, Data curation. **Leighton Barnden:** Writing – review & editing, Supervision, Funding acquisition.

Data sharing statement

Unidentified data can be made available upon request.

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Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbih.2025.101142.

Data availability

Data will be made available on request.

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